



Multicomponent, Multiphase Thermodynamics of Swelling Porous Media with Electroquasistatics:

II. Constitutive Theory

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Abstract. In Part I macroscopic field equations of mass, linear and angular momentum, energy, and the quasistatic form of Maxwell's equations for a multiphase, multicomponent medium were derived. Here we exploit the entropy inequality to obtain restrictions on constitutive relations at the macroscale for a 2-phase, multiple-constituent, polarizable mixture of fluids and solids. Specific emphasis is placed on charged porous media in the presence of electrolytes. The governing equations for the stress tensors of each phase, flow of the fluid through a deforming medium, and diffusion of constituents through such a medium are derived. The results have applications in swelling clays (smectites), biopolymers, biological membranes, pulsed electrophoresis, chromatography, drug delivery, and other swelling systems.

Key words: mixture theory, electrodynamics, swelling, constitutive equations.

1. Introduction

We continue our investigation into the form of the governing equations for a multiple-component, multiple-phase, polarizable, swelling porous medium with charged particles subject to an electric field. In Part I of this series, we derived the macroscopic field equations in which it is the total electric field which affects the species' conservation of momentum and energy. In this paper, we exploit the entropy inequality in the sense of Coleman and Noll (1963) in order to obtain restrictions on constitutive equations in terms of *macroscopic* variables. This approach differs from classical averaging (Whitaker, 1967, 1969; Plumb and Whitaker, 1996; del Rio and Whitaker, 2000a, b) or homogenization (Bensoussan *et al.*, 1978; Sanchez-Palencia, 1980; Douglas and Arbogast, 1990) where both the field equations and microscopic constitutive equations are upscaled to obtain macroscale equations in terms of microscale geometry and microscale constitutive coefficients. The advantage of homogenization and averaging is that *if* one knows the microscale

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constitutive equations and geometry, then one can obtain to first-order (and second-order...) constitutive equations (including the coefficients) at the macroscale in terms of microscopic variables. The HMT approach assumes only the variables upon which constitutive variables may depend, and produces restrictions of the form of constitutive equations in terms of *macroscopic* variables. HMT does not assume any microscale constitutive relations nor any specific microscale geometry, which in such complex media as considered here are not satisfactorily known. This point is of significant importance for natural media such as geophysical environs or polymeric systems, which are never well-characterized geometrically at the microscale.

For simplicity we consider only a liquid–solid system. The mixture may be charge-neutral, although if it is considered to be charge-neutral neither the phases nor species face this requirement individually. We assume that interfacial properties such as excess mass density, free charge on interfaces, and interface currents are negligible; although the present theory can be extended to explicitly incorporate these effects (Gray and Hassanizadeh, 1991). However, it should be noted that for many practical materials where the ratio of surface area to volume of the solid phase remains constant, interfacial characteristics can be accounted for in this theory. This is because in the case where surface to volume ratio is constant, charge per unit mass of the solid phase is directly proportional to charge per unit surface area. Likewise the theory presented here also applies to materials in which the charge of the solid itself changes. This theory would *not* apply to materials in which the charge of the solid phase and of the interface vary independently. So, for example, the theory applies to clay minerals, which has a charge associated only with their surfaces, and to polymers which have a charge associated with the polymer – with polymer surfaces often not well-defined.

In deriving restrictions on the form of the constitutive equations, we follow (Eringen, 1967, 1998) and view fluxes as constitutive. In earlier work, Eringen (1998) considered constituent electric fields throughout for a non-swelling porous medium; Huyghe and Janssen (1999) considered a single electric field in a deformable two-phase porous medium with no exchange terms in the conservation equations; and Gu et al. (1999) considered a charged swelling medium with no electric field.

2. Constitutive Assumptions and the Entropy Inequality

The full entropy inequality which is exploited in subsequent sections is presented in Appendix B. The basic notation is found in Part I so only new terms are defined herein. We assume the medium consists of a liquid phase (denoted by $\alpha = l$), and a solid phase, (denoted by $\alpha = s$). We assume that entropy generation must be non-negative for the total body, that is,

$$\rho \hat{\Lambda} = \sum_{\alpha} \sum_j \varepsilon^{\alpha} \rho^{\alpha_j} \hat{\Lambda}^{\alpha_j} \geq 0. \quad (1)$$

Further we assume a form of local equilibrium wherein there is one temperature for all constituents and all phases, that is, $T^{\alpha j}(x, t) = T(x, t)$ for all constituents j and all phases α . This effectively states that the rate of heat transfer between constituents is much faster than the time scales of interest to the problem.

To couple the entropy and energy equations, it is necessary to relate the fluxes and sources of entropy to the fluxes and sources of heat. We assume the processes are simple in the sense of Eringen (1967). In this sense several possible relations are admissible. Among these are

$$\phi^{\alpha j} = \frac{q^{\alpha j}}{T}, \quad b^{\alpha j} = \frac{h^{\alpha j}}{T}, \quad (2)$$

$$\phi^{\alpha} = \frac{q^{\alpha}}{T}, \quad b^{\alpha} = \frac{h^{\alpha}}{T}, \quad (3)$$

$$\sum_{\alpha} \varepsilon^{\alpha} \phi^{\alpha} = \sum_{\alpha} \frac{\varepsilon^{\alpha} q^{\alpha}}{T}, \quad \sum_{\alpha} \varepsilon^{\alpha} b^{\alpha} = \sum_{\alpha} \frac{\varepsilon^{\alpha} h^{\alpha}}{T}. \quad (4)$$

The expressions in (4) have been used in Eringen (1998), and the relations in (2) have been used in Benach and Müller (1973), Hassanizadeh and Gray (1980), Achanta *et al.* (1994), Bennethum and Cushman (1996b). More general expressions which simplify to the above are used in extended thermodynamics (Müller and Ruggeri, 1993; Jou *et al.*, 1996). The question is whether the processes governing the behavior of the constituents themselves, the individual phases, or the bulk material are simple. The assumption of any one of these does not imply any other due to microscale/macroscale relationships between heat fluxes and heat sources (Bennethum and Cushman, 2002). This problem is complicated further because the macroscale definition of the heat flux depends on how one incorporates microscale fluctuations (see Part I, Bennethum and Cushman, 2002). The differences manifest themselves explicitly in the constitutive relations obtained for diffusive fluxes and the chemical potential. For example, if the relations in (4) are used then it can be shown that the chemical potentials of two ‘different’ species at equilibrium must be equal, which is inconsistent with Gibbsian thermostatics (Callen, 1985). To the authors’ knowledge, the relations in (2) do not result in any physical inconsistencies in near-equilibrium processes, and these are the relations used herein.

Define the Helmholtz free energy for the species and the internal Helmholtz free energy for the bulk phase as

$$A^{\alpha j} = e^{\alpha j} - T \eta^{\alpha j}, \quad A^{\alpha} = \sum_{j=1}^N C^{\alpha j} A^{\alpha j} \quad (5)$$

and introduce a modified Helmholtz free energy as

$$\tilde{A}^{\alpha j} = e^{\alpha j} - T \eta^{\alpha j} - \frac{1}{\rho^{\alpha j}} \mathbf{E}_T \cdot \mathbf{P}^{\alpha j}, \quad \tilde{A}^{\alpha} = \sum_{j=1}^N C^{\alpha j} \tilde{A}^{\alpha j}, \quad (6)$$

where $C^{\alpha j}$ is the mass fraction of constituent j in phase α given by $C^{\alpha j} = \rho^{\alpha j} / \rho^\alpha$. The purpose of introducing the modified Helmholtz potential is to reduce the amount of manipulations required to obtain the entropy inequality, as either $\mathbf{P}^{\alpha j}$ or $\mathbf{E}_T$ must be constitutive (dependent) variables.

To obtain the entropy inequality, begin with Equation (1) and the entropy balance (66) of Part I. Eliminate $\phi^{\alpha j}$ and $b^{\alpha j}$ using (2); $h^{\alpha j}$ using the energy balance Equation (60) of Part I; and $\hat{Q}^{\alpha j}$, $\hat{Q}_\beta^{\alpha j}$, $\hat{\eta}^{\alpha j}$, $\hat{\Phi}_\beta^{\alpha j}$ using (62), (63), (68), and (69), respectively, of Part I. This gives

$$\begin{aligned} & \sum_{\alpha} \varepsilon^{\alpha} \rho^{\alpha} T \hat{\Lambda}^{\alpha} \\ &= \sum_{\alpha=l,s} \sum_{j=1}^N \left[-\varepsilon^{\alpha} \rho^{\alpha j} \frac{D^{\alpha j} \tilde{A}^{\alpha j}}{Dt} - \varepsilon^{\alpha} \rho^{\alpha j} \eta^{\alpha j} \frac{D^{\alpha j} T}{Dt} + \right. \\ & \quad + \frac{\varepsilon^{\alpha}}{\rho^{\alpha j}} \mathbf{E}_T \cdot \mathbf{P}^{\alpha j} \frac{D^{\alpha j} \rho^{\alpha j}}{Dt} - \varepsilon^{\alpha} \mathbf{P}^{\alpha j} \cdot \frac{D^{\alpha j} \mathbf{E}_T}{Dt} - \varepsilon^{\alpha} \mathbf{E}_T \cdot \frac{D^{\alpha j} \mathbf{P}^{\alpha j}}{Dt} + \\ & \quad + \frac{\varepsilon^{\alpha}}{T} \mathbf{q}^{\alpha j} \cdot \nabla T + \varepsilon^{\alpha} \mathbf{t}^{\alpha j} : \nabla \mathbf{v}^{\alpha j} + \varepsilon^{\alpha} \mathcal{J}^{\alpha j} \cdot \mathbf{E}_T + \\ & \quad + \frac{\partial(\varepsilon^{\alpha} \mathbf{P}^{\alpha j})}{\partial t} \cdot \mathbf{E}_T + \nabla \cdot (\varepsilon^{\alpha} \mathbf{v}^{\alpha j} \mathbf{P}^{\alpha j}) \cdot \mathbf{E}_T + \frac{1}{2} \varepsilon_0 \mathbf{E}_T \cdot \mathbf{E}^{\alpha j} \frac{D^{\alpha j} \varepsilon^{\alpha}}{Dt} + \\ & \quad + \varepsilon^{\alpha} \rho^{\alpha j} T \left(\sum_{\beta \neq \alpha} \tilde{e}_\beta^{\alpha j} \eta^{\alpha j} + \hat{r}^{\alpha j} \eta^{\alpha j} \right) - \varepsilon^{\alpha} \rho^{\alpha j} \left(\sum_{\beta \neq \alpha} \hat{T}_\beta^{\alpha j} \cdot \mathbf{v}^{\alpha j} + \right. \\ & \quad + \tilde{e}_\beta^{\alpha j} \left(\tilde{A}^{\alpha j} + T \eta^{\alpha j} + \frac{1}{\rho^{\alpha j}} \mathbf{E}_T \cdot \mathbf{P}^{\alpha j} + \frac{1}{2} (\mathbf{v}^{\alpha j})^2 \right) \left. \right) - \varepsilon^{\alpha} \rho^{\alpha j} \hat{i}^{\alpha j} \cdot \mathbf{v}^{\alpha j} - \\ & \quad \left. - \varepsilon^{\alpha} \rho^{\alpha j} \hat{r}^{\alpha j} \left(\tilde{A}^{\alpha j} + T \eta^{\alpha j} + \frac{1}{\rho^{\alpha j}} \mathbf{E}_T \cdot \mathbf{P}^{\alpha j} + \frac{1}{2} (\mathbf{v}^{\alpha j})^2 \right) \right] \geq 0, \end{aligned}$$

where the contraction operator $A : B$ is, in indicial notation, $A_{ij} B_{ij}$. Next sum on j and write $\mathbf{q}^{\alpha j} \cdot \nabla T$, $\eta^{\alpha j} (D^{\alpha j} T / Dt)$, $\mathbf{t}^{\alpha j} : \nabla \mathbf{v}^{\alpha j}$, in terms of $\mathbf{q}^{\alpha} \cdot \nabla T$, $\eta^{\alpha} (D^{\alpha} T / Dt)$, $\mathbf{t}^{\alpha} : \nabla \mathbf{v}^{\alpha}$, respectively, using the relations given in Appendix B of Part I. Also using

$$\begin{aligned} & - \sum_{j=1}^N \varepsilon^{\alpha} \rho^{\alpha j} \frac{D^{\alpha j} \tilde{A}^{\alpha j}}{Dt} \\ &= -\varepsilon^{\alpha} \rho^{\alpha} \frac{D^{\alpha} \tilde{A}^{\alpha}}{Dt} - \tilde{A}^{\alpha} \frac{D^{\alpha} (\varepsilon^{\alpha} \rho^{\alpha})}{Dt} + \sum_{j=1}^N \left[\tilde{A}^{\alpha j} \frac{D^s (\varepsilon^{\alpha} \rho^{\alpha j})}{Dt} + \right. \\ & \quad \left. + \tilde{A}^{\alpha j} \mathbf{v}^{\alpha j, \alpha} \cdot \nabla (\varepsilon^{\alpha} \rho^{\alpha j}) + \tilde{A}^{\alpha j} \mathbf{v}^{\alpha, s} \cdot \nabla (\varepsilon^{\alpha} \rho^{\alpha j}) - \mathbf{v}^{\alpha j, \alpha} \cdot \nabla (\varepsilon^{\alpha} \rho^{\alpha j} \tilde{A}^{\alpha j}) \right] \end{aligned}$$

$$\begin{aligned}
&= -\varepsilon^\alpha \rho^\alpha \frac{D^\alpha \tilde{A}^\alpha}{Dt} + \sum_{j=1}^N \left[\tilde{A}^{\alpha_j} \frac{D^s(\varepsilon^\alpha \rho^{\alpha_j})}{Dt} - \varepsilon^\alpha \rho^{\alpha_j} \mathbf{v}^{\alpha_j, \alpha} \cdot \nabla \tilde{A}^{\alpha_j} + \right. \\
&\quad \left. + \varepsilon^\alpha \rho^{\alpha_j} \tilde{A}^{\alpha_j} \nabla \cdot \mathbf{v}^\alpha + \tilde{A}^{\alpha_j} \mathbf{v}^{\alpha, s} \cdot \nabla(\varepsilon^\alpha \rho^{\alpha_j}) - \sum_{\beta \neq \alpha} \varepsilon^\alpha \rho^{\alpha_j} \tilde{A}^{\alpha_j} \tilde{\mathbf{e}}_\beta^{\alpha_j} \right],
\end{aligned}$$

where a comma in the superscript denotes difference (e.g. $\mathbf{v}^{\alpha, s} = \mathbf{v}^\alpha - \mathbf{v}^s$), and putting all time derivatives in terms of D^s/Dt using

$$\frac{D^{\alpha_j}}{Dt} = \frac{D^s}{Dt} + \mathbf{v}^{\alpha_j, \alpha} \cdot \nabla + \mathbf{v}^{\alpha, s} \cdot \nabla$$

yields the entropy inequality before any constitutive assumptions have been made

$$\begin{aligned}
&\sum_{\alpha} \varepsilon^\alpha \rho^\alpha T \hat{\Lambda}^\alpha \\
&= - \sum_{\alpha} \varepsilon^\alpha \rho^\alpha \left(\frac{D^\alpha \tilde{A}^\alpha}{Dt} + \eta^\alpha \frac{D^\alpha T}{Dt} \right) + \sum_{\alpha} \varepsilon^\alpha \mathcal{J}^\alpha \cdot \mathbf{E}_T + \\
&\quad + \sum_{\alpha} \left[\varepsilon^\alpha \mathbf{t}^\alpha + \varepsilon^\alpha \mathbf{P}^\alpha \cdot \mathbf{E}_T \mathbf{I} + \sum_{j=1}^N \varepsilon^\alpha \rho^{\alpha_j} \left(\tilde{A}^{\alpha_j} \mathbf{I} + \mathbf{v}^{\alpha_j, \alpha} \mathbf{v}^{\alpha_j, \alpha} \right) \right] : \nabla \mathbf{v}^\alpha + \\
&\quad + \sum_{\alpha} \sum_{j=1}^N \left[\varepsilon^\alpha \mathbf{t}^{\alpha_j} + \varepsilon^\alpha \mathbf{P}^{\alpha_j} \cdot \mathbf{E}_T \mathbf{I} \right] : \nabla \mathbf{v}^{\alpha_j, \alpha} + \\
&\quad + \sum_{\alpha} \dot{\varepsilon}^\alpha \left[\frac{1}{2} \varepsilon_o \mathbf{E}_T \cdot \mathbf{E}^\alpha + \mathbf{E}_T \cdot \mathbf{P}^\alpha + \rho^\alpha A^\alpha \right] + \sum_{\alpha} \frac{\varepsilon^\alpha}{T} \nabla T \times \\
&\quad \times \left\{ \mathbf{q}^\alpha + \sum_{j=1}^N \left[\rho^{\alpha_j} \mathbf{v}^{\alpha_j, \alpha} \left(\tilde{A}^{\alpha_j} + \frac{1}{2} \mathbf{v}^{\alpha_j, \alpha} \cdot \mathbf{v}^{\alpha_j, \alpha} \right) - \mathbf{t}^{\alpha_j} \cdot \mathbf{v}^{\alpha_j, \alpha} \right] \right\} + \\
&\quad + \sum_{\alpha} \sum_{j=1}^N \varepsilon^\alpha \frac{D^s \rho^{\alpha_j}}{Dt} \left[\frac{1}{\rho^{\alpha_j}} \mathbf{E}_T \cdot \mathbf{P}^{\alpha_j} + A^{\alpha_j} \right] - \sum_{\alpha} \varepsilon^\alpha \dot{\mathbf{E}}_T \cdot \mathbf{P}^{\alpha_j} + \\
&\quad + \mathbf{v}^{l, s} \cdot \left[-\varepsilon^l \rho^l \hat{\mathbf{T}}_s^l + \sum_{j=1}^N \left(A^{l_j} \nabla(\varepsilon^l \rho^{l_j}) + \frac{1}{\rho^{l_j}} \mathbf{E}_T \cdot \mathbf{P}^{l_j} \nabla(\varepsilon^l \rho^{l_j}) - \right. \right. \\
&\quad \left. \left. - \varepsilon^l \nabla \mathbf{E}_T \cdot \mathbf{P}^{l_j} + \frac{1}{2} \varepsilon_o \mathbf{E}_T \cdot \mathbf{E}^{l_j} \nabla \varepsilon^l \right) \right] + \sum_{\alpha} \sum_{j=1}^N \mathbf{v}^{\alpha_j, \alpha} \times \\
&\quad \times \left[-\varepsilon^\alpha \rho^{\alpha_j} \left(\hat{\mathbf{i}}^{\alpha_j} + \sum_{\beta \neq \alpha} \hat{\mathbf{T}}_\beta^{\alpha_j} \right) - \varepsilon^\alpha \rho^{\alpha_j} \nabla \tilde{A}^{\alpha_j} - \varepsilon^\alpha q_e^{\alpha_j} \mathbf{E}_T + \right. \\
&\quad \left. + \frac{1}{\rho^{\alpha_j}} \mathbf{E}_T \cdot \mathbf{P}^{\alpha_j} \nabla(\varepsilon^\alpha \rho^{\alpha_j}) - \varepsilon^\alpha \nabla \mathbf{E}_T \cdot \mathbf{P}^{\alpha_j} + \frac{1}{2} \varepsilon_o \mathbf{E}_T \cdot \mathbf{E}^{\alpha_j} \nabla \varepsilon^\alpha \right] -
\end{aligned}$$

$$\begin{aligned}
& - \sum_{\alpha} \sum_{j=1}^N \sum_{\beta \neq \alpha} \varepsilon^{\alpha} \rho^{\alpha_j} \tilde{e}_{\beta}^{\alpha_j} \left[\frac{1}{\rho^{\alpha_j}} \mathbf{E}_T \cdot \mathbf{P}^{\alpha_j} + \tilde{A}^{\alpha_j} + \right. \\
& + \tilde{A}^{\alpha} + \frac{1}{2} (\mathbf{v}^{\alpha, s})^2 + \frac{1}{2} (\mathbf{v}^{\alpha_j, \alpha})^2 \left. \right] + \\
& + \sum_{\alpha} \sum_{j=1}^N \varepsilon^{\alpha} \rho^{\alpha_j} \tilde{r}^{\alpha_j} \left[-\frac{1}{2} (\mathbf{v}^{\alpha_j, \alpha})^2 - \tilde{A}^{\alpha_j} - \frac{1}{\rho^{\alpha_j}} \mathbf{E}_T \cdot \mathbf{P}^{\alpha_j} \right] \geq 0, \quad (7)
\end{aligned}$$

where a superimposed dot denotes the material time derivative with respect to the solid phase (e.g. $\dot{\varepsilon}^{\alpha} = \partial \varepsilon^{\alpha} / \partial t + \mathbf{v}^s \cdot \nabla \varepsilon^{\alpha}$), and $\mathbf{I}$ is the identity matrix. All symbols not defined explicitly in this paper are defined in Part I.

2.1. CHARGE NEUTRALITY

Charge neutrality is an assumption often made when no external electric field is applied (Gu *et al.*, 1999; Huyghe and Janssen, 1999). Assuming a charge-neutral system implies that the time scale at which charge imbalances may occur is small compared to the time scale involved with other processes.

In this manuscript charge neutrality is viewed in the same manner as enforcing incompressibility of a solid phase. For a pure solid phase the independent variables usually include strain, temperature, . . . but does not include density (due to the dependence of density and strain through the continuity equation for a non-reactive solid). In this case there is an additional unknown (the Lagrange Multiplier), which enforces the relationship between the various strain elements resulting from the incompressible form of the continuity Equation (Eringen, 1967). A priori one does not know whether the solid can be treated as an incompressible solid or not – this depends on the material properties and the particular dynamics of the situation. Thus, exploiting the entropy inequality using the incompressibility assumption is a more general form: for a compressible medium, one just drops the Lagrange multiplier which enforces the relationship between the strain elements. Note that in this case there has been an additional constraint (the material time derivative of the density must be zero) and an additional unknown (the Lagrange multiplier).

Likewise the electrical neutrality condition cannot be made until the particular material and dynamic process under which it goes is known. Permittivity and conductance of the material are properties which have to be experimentally determined, just as compressibility is for the solid phase. Enforcing the charge neutrality condition using the Lagrange multiplier means that the charge densities of all materials cannot vary independently, and has nothing to do with Maxwell's equations a priori. In this case, the additional equation is the material time rate of change of the total charge and the additional unknown is the Lagrange multiplier. If the problem is not charge neutral, then the Lagrange multiplier enforcing charge neutrality should be dropped, and no other changes need be made.

The form of the charge-neutrality condition is also made based on the analogy with an incompressible solid. The constraint for an incompressible solid phase can be expressed in integral form as follows:

$$\frac{D}{Dt} \int_{V(\mathbf{x})} \rho \, dV = 0,$$

or that the material time rate of change of the total mass in a *material* framework must be zero. If one uses the Eulerian framework, one has the continuity equation in its entirety. The local form of the incompressibility condition is obtained by assuming that the above relation holds for all material volumes and that the density function is continuous. Applying the same reasoning to the total charge, $q_e = \varepsilon^l q_e^l + \varepsilon^s q_e^s$, we have

$$\frac{D^s}{Dt} \int_{V(\mathbf{x})} q_e \, dV = 0,$$

or $(D^s/Dt)(q_e) = \mathbf{0}$, which is the form we enforce using a Lagrange multiplier.

In addition to charge neutrality, we enforce many of the balance laws weakly using the Lagrange multiplier approach (Liu, 1972). The equations and their associated Lagrange multipliers are listed below.

Lagrange Mult.	Equation from (Bennethum and Cushman, 2002)	Lagrange Mult.	Equation from (Bennethum and Cushman, 2002)
$\lambda_\rho^{\alpha_j}$	Continuity Equation α_j , (15)	$\lambda_D^{\alpha_j}$	Gauss' law (22)
$\Lambda_E^{\alpha_j}$	Faraday's Law (28)	$\lambda_{q_e}^{\alpha_j}$	Conserv. of charge (44)
Λ	$(D^s/Dt)(\varepsilon^l q_e^l + \varepsilon^s q_e^s) = 0$		

Ampère's law is derivable from the conservation of charge and Gauss' law and so is not enforced directly. Scale at which imbalances may occur is small compared to the time scale involved with other processes.

2.2. INDEPENDENT VARIABLES

The unknowns in this system include

$$\varepsilon^l, \rho^{\alpha_j}, \mathbf{v}^{\alpha_j}, T, \mathbf{E}^{\alpha_j}, q_e^{\alpha_j}, \quad (8)$$

$$\widehat{\mathbf{e}}_\beta^{\alpha_j}, \widehat{\mathbf{r}}^{\alpha_j}, \mathbf{t}^{\alpha_j}, \mathbf{t}^\alpha, \widehat{\mathbf{T}}_\beta^{\alpha_j}, \widehat{\mathbf{i}}^{\alpha_j}, \quad (9)$$

$$\widehat{\mathbf{m}}^{\alpha_j}, \widehat{\mathbf{M}}^{\alpha_j}, \widehat{\mathbf{m}}_\beta^{\alpha_j}, A^{\alpha_j}, \mathbf{q}^{\alpha_j}, \eta^{\alpha_j}, \widehat{\Phi}_\beta^{\alpha_j}, \widehat{\eta}^{\alpha_j}, \quad (10)$$

$$\mathbf{P}^{\alpha_j}, \widehat{\mathbf{d}}^{\alpha_j}, \widehat{\mathbf{d}}_\beta^{\alpha_j}, \widehat{\sigma}^{\alpha_j}, \widehat{\sigma}_\beta^{\alpha_j}, \mathcal{J}^{\alpha_j}, \widehat{q}^{\alpha_j}, \widehat{Z}_\beta^{\alpha_j}, \quad j = 1, \dots, N. \quad (11)$$

The variables in the first row, (8), are the primary unknowns. The remaining variables, (9–11), are considered constitutive and are a function of *constitutive independent variables*. In order to close the system, one additional equation is needed, which corresponds to the unknown ε^l . This is known as the closure problem, and it arises from the homogenization of the microscopic geometry. To close the system we follow (Bowen, 1976; Gray and Hassanizadeh, 1991; Achanta *et al.*, 1994) and view the time rate of change of the volume fraction $D\varepsilon^l/Dt$ as a constitutive variable.

The choice of constitutive independent variables is made based on knowledge of the system being modeled. Here we assume the fluid may behave as a Newtonian fluid and the solid as an elastic solid, hence we include the rate of deformation tensor, $\mathbf{d}^l$, and the strain tensor, $\mathcal{E}^s$. Since the solid phase may be disconnected, the macroscale strain tensor is not the average of the microscale strain, but is defined in terms of the deformation gradient $\mathbf{F} = \nabla_o \mathbf{x}$, $\mathcal{E}^s = \frac{1}{2}(\mathbf{F}_s^T \mathbf{F}_s - \mathbf{I})$, where ∇_o denotes differentiation with respect to the macroscopic material particle. Thus the strain tensor is a measure of the geometry of the solid phase. Further, we are particularly interested in modeling materials in which the solid and fluid phases have electrochemical interactions, so that the behavior of the liquid phase may strongly depend upon its proximity to the solid phase. Thus we incorporate the volume fraction, ε^l , as an independent variable. The independent variables which are used to define the constitutive variables include

$$\begin{aligned} \varepsilon^l, T, \rho^{\alpha_j}, \mathbf{v}^{l,s}, \mathbf{v}^{\alpha_j,\alpha}, \mathcal{E}^s, \mathbf{E}_T, z^{\alpha_j}, \\ \nabla \varepsilon^l, \nabla T, \nabla \rho^{\alpha_j}, \mathbf{d}^l, \boldsymbol{\omega}^\alpha, \nabla \mathbf{v}^{l,j,l}, \nabla \mathcal{E}^s, \nabla \mathbf{E}_T, \quad j = 1, \dots, N, \\ \alpha = l, s \end{aligned} \quad (12)$$

where because the liquid phase may be polarizable and may depend strongly on the geometry of the solid phase (Bennethum and Cushman, 2002), we have also included the vorticity tensor, $\boldsymbol{\omega}^l = (\nabla \mathbf{v} - (\nabla \mathbf{v})^T)$. Note that we have incorporated the total electric field as an independent variable, as opposed to the electric fields of each constituent or each phase. This is because it is assumed that all constitutive variables are measured with respect to the total electric field.

The charge of a species, z^{α_j} , has units of charge of α_j per unit mass of α_j . This results in terms such as $\partial A^\alpha / \partial z^{\alpha_j}$ which are evaluated holding all other independent variables in (12), such as volume fraction and densities, fixed. This allows for disassociation of ions, or changes in the charge density of the solid phase, either through the change in surface charge density or through charge density of the bulk phase, but not both (see the introduction).

To simplify the results, we relax the Principle of Equipresence (Truesdell and Noll, 1965) and assume that the modified Helmholtz potential energies $\tilde{A}^\alpha$ are a function of a subset of the above constitutive independent variables

$$\tilde{A}^l = \tilde{A}^l(\varepsilon^l, T, \rho^{l,j}, \mathbf{v}^{l,s}, \mathbf{v}^{l,j,l}, \mathcal{E}^s, \mathbf{E}_T, z^{l,j}, \nabla T, \nabla \rho^{l,j}, \mathbf{d}^l, \boldsymbol{\omega}^l, \nabla \mathbf{v}^{l,j,l}) \quad (13)$$

$$\tilde{A}^s = \tilde{A}^s(\varepsilon^l, T, \rho^{sj}, \mathbf{v}^{sj,s}, \boldsymbol{\varepsilon}^s, \mathbf{E}_T, z^{sj}, \nabla T, \nabla \rho^{sj}, \boldsymbol{\omega}^s). \quad (14)$$

We note that including the additional independent variables does not change the results if one modifies the definitions of pressure, chemical potential, etc. see e.g., Bennethum and Cushman (1996b). The entropy inequality is now expanded in the traditional manner (Eringen, 1967; Hassanizadeh and Gray, 1979; Bennethum *et al.*, 2000) and is presented in Appendix B.

3. Nonequilibrium Constitutive Restrictions

The following variables are neither constitutive nor independent:

$$\frac{D^s \rho^{\alpha j}}{Dt}, \frac{D^s z^{\alpha j}}{Dt}, \dot{T}, \dot{\mathbf{E}}_T, \mathbf{d}^s, \nabla \mathbf{v}^{sj,s}, \nabla \dot{T}, \quad (15)$$

$$\dot{\mathbf{v}}^{\alpha j, \alpha}, \nabla \left(\frac{D^s \rho^{\alpha j}}{Dt} \right), \dot{\mathbf{v}}^{l,s}, \dot{\mathbf{d}}^l, \dot{\boldsymbol{\omega}}^l, \nabla \dot{\mathbf{v}}^{lj,l}, \quad (16)$$

$$\nabla \times \mathbf{E}^{\alpha j}, \nabla \cdot \mathbf{E}^{\alpha j}, \quad (17)$$

where $j = 1, \dots, N$ for all variables not containing $\mathbf{v}^{\alpha j, \alpha}$, since $\sum_{j=1}^N \rho^{\alpha j} \mathbf{v}^{\alpha j, \alpha} = \mathbf{0}$. Thus for example, $\mathbf{v}^{sj,s}$ is indexed from $j = 1, \dots, N-1$ in order to keep the list of variables functionally independent. Since these terms appear linearly in the entropy inequality, their coefficients must be zero. This results in the following restrictions (corresponding directly with the terms in (15)):

$$\lambda_{\rho}^{\alpha j} = \rho^{\alpha} \frac{\partial \tilde{A}^{\alpha}}{\partial \rho^{\alpha j}} - \tilde{A}^{\alpha j} - \frac{1}{\rho^{\alpha j}} \mathbf{E}_T \cdot \mathbf{P}^{\alpha j} - z^{\alpha j} \Lambda, \quad j = 1, \dots, N, \quad (18)$$

$$\lambda_{q_e}^{\alpha j} + \Lambda = \frac{\rho^{\alpha}}{\rho^{\alpha j}} \frac{\partial \tilde{A}^{\alpha}}{\partial z^{\alpha j}}, \quad (19)$$

$$\sum_{\alpha=l,s} \varepsilon^{\alpha} \rho^{\alpha} \left(\eta^{\alpha} + \frac{\partial \tilde{A}^{\alpha}}{\partial T} \right) = 0, \quad (20)$$

$$\sum_{\alpha} \varepsilon^{\alpha} \mathbf{P}^{\alpha} = - \sum_{\alpha} \varepsilon^{\alpha} \rho^{\alpha} \frac{\partial \tilde{A}^{\alpha}}{\partial \mathbf{E}_T}, \quad (21)$$

$$\begin{aligned} \varepsilon^s \mathbf{t}_{\text{sym}}^s = & - \sum_{j=1}^N \varepsilon^s \rho^{sj} (\lambda_{\rho}^{sj} + \tilde{A}^{sj}) + \varepsilon^s \mathbf{t}_e^s + \varepsilon^l \mathbf{t}_s^l - \varepsilon^s \mathbf{P}^s \cdot \mathbf{E}_T \mathbf{I} - \\ & - \sum_{j=1}^N \varepsilon^s \rho^{sj} \mathbf{v}^{sj,s} \mathbf{v}^{sj,s}, \end{aligned} \quad (22)$$

$$\mathbf{t}^{sj} - \frac{\rho^{sj}}{\rho^{sN}} \mathbf{t}^{sN} = - \left(\mathbf{P}^{sj} \cdot \mathbf{E}_T - \frac{\rho^{sj}}{\rho^{sN}} \mathbf{P}^{sN} \cdot \mathbf{E}_T \right) \mathbf{I} - \rho^{sj} (\lambda_{\rho}^{sj} - \lambda_{\rho}^{sN}) \mathbf{I}, \quad (23)$$

$$\varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \nabla T} + \varepsilon^s \rho^s \frac{\partial \tilde{A}^s}{\partial \nabla T} = 0, \quad (24)$$

where we have defined the effective stress tensor and hydration stress tensor as

$$\mathbf{t}_e^s = \rho^s \nabla \mathbf{F}^s \cdot \frac{\partial \tilde{A}^s}{\partial \mathcal{E}^s} \cdot (\nabla \mathbf{F}^s)^T, \quad \mathbf{t}_s^l = \rho^l \nabla \mathbf{F}^s \cdot \frac{\partial \tilde{A}^l}{\partial \mathcal{E}^s} \cdot (\nabla \mathbf{F}^s)^T, \quad (25)$$

respectively. The restrictions obtained from the coefficients of the variables listed in (16) indicate that the modified Helmholtz free energies are not a function of $\mathbf{v}^{\alpha_j, \alpha}$, $\nabla \rho^{\alpha_j}$, $\mathbf{v}^{l, s}$, $\mathbf{d}^l$, $\boldsymbol{\omega}^\alpha$, $\nabla \mathbf{v}^{l, l}$, and the restrictions corresponding to the variables listed in (17) require that the Lagrange multipliers, $\lambda_{q_e}^{\alpha_j}$, $\Lambda_E^{\alpha_j}$, and $\lambda_D^{\alpha_j}$ must all be identically zero.

Equations (18) and (19) define the Lagrange multipliers $\lambda_\rho^{\alpha_j}$ and $\lambda_{q_e}^{\alpha_j}$, respectively. Equation (20) states that η^α and T are dual variables with respect to the *modified* Helmholtz potential, and this is in agreement with Benach and Müller (1973) and Eringen (1998). Equation (24) states that the modified Helmholtz potential of the entire system is independent of ∇T .

The definitions of the effective stress tensor indicates that the effective stress tensor measures a change in energy of the solid phase with respect to the strain in the solid phase. The strain in the solid phase is purely a geometrical quantity (intuitively, one takes the solid phase, smears it out to obtain a continuous medium at the macroscale, and measures its deformation – see the definition of $\mathcal{E}$ following (11)). If the solid phase is connected, the strain represents an average of the microscale strain, and the effective stress tensor is non-negligible if strain occurs. For a disconnected solid phase, the energy of the solid phase is negligible for large movements of particles, and hence the effective stress tensor is negligible. However, in the case where the medium swells (high interaction between liquid and solid phase), and where the solid phase is disconnected, the hydration stress tensor measures the stress the liquid phase supports when the solid phase is sheared. This can occur at low moisture content, see, for example, Cushman (1990), so that in this case although the effective stress tensor is negligible, the hydration stress tensor is not. Most swelling porous media have a structure such that both the solid and liquid phase can support shear, in which case neither the hydration nor the effective stress tensors can be neglected. See Bennethum *et al.* (1997) for a more detailed discussion on this topic.

Equation (21) implies that it is the polarization of the entire medium which is dual to the electric field in this representation. This is in contrast with Eringen (1998) in which it is shown that the polarization of each component is dual to the constituent's electric field. The disparity results from our choice of independent variables – we use $\mathbf{E}_T$ and Eringen uses $\mathbf{E}^{\alpha_j}$ as independent variables.

We define the pressure thermodynamically by

$$p^\alpha = \sum_{j=1}^N \rho^{\alpha_j} (\lambda_\rho^{\alpha_j} + \tilde{A}^{\alpha_j}) + \mathbf{P}^\alpha \cdot \mathbf{E}_T + q_e^\alpha \Lambda \quad (26)$$

$$= \sum_{j=1}^N \left(\rho^\alpha \rho^{\alpha_j} \frac{\partial \tilde{A}^\alpha}{\partial \rho^{\alpha_j}} \right). \quad (27)$$

To see how this variable corresponds to what is classically thought of as pressure, we rewrite (22), the expression for the symmetric part of the solid-phase stress tensor, as

$$\mathbf{t}_{\text{sym}}^s = -(p^s - q_e^s \Lambda) \mathbf{I} + \mathbf{t}_e^s + \frac{\varepsilon^l}{\varepsilon^s} \mathbf{t}_s^l - \sum_{j=1}^N \rho^{s_j} \mathbf{v}^{s_j, s} \mathbf{v}^{s_j, s} \quad (28)$$

so that by comparison (Eringen, 1967), we see that $p^s - q_e^s \Lambda$ represents what is traditionally thought of as pressure. Charges and the electric field enter into this expression through the term containing the Lagrange multiplier enforcing charge neutrality, $q_e^s \Lambda$, and through the definition of $\tilde{A}^\alpha$ in the remaining terms. The role played by the effective and hydration stress tensors are discussed in Bennethum *et al.* (1996, 1997), Murad *et al.* (1995), and Murad and Cushman (1996). We remark that if the solid phase is considered incompressible then the material time derivative of ρ^{s_j} is zero, ρ^{s_j} is not considered an independent variable, and $\lambda_\rho^{s_j}$ is the Lagrange multiplier which enforces the remaining part of the conservation of mass for constituent s_j . In that case we obtain the same results, except that p^s is a primary unknown which must be solved for directly.

We define the chemical potential as the change of the total Helmholtz potential with respect to the mass of species j (keeping the total volume fixed) (Callen, 1985). Intuitively, it is the scalar quantity representing the chemical energy required to insert that species into the mixture. Formally dividing this thermodynamic definition by the volume of the REV, we have

$$\mu^{\alpha_j} = \frac{\partial(\varepsilon^\alpha \rho^\alpha \tilde{A}^\alpha)}{\partial(\varepsilon^\alpha \rho^{\alpha_j})} = \tilde{A}^\alpha + \rho^\alpha \left. \frac{\partial \tilde{A}^\alpha}{\partial \rho^{\alpha_j}} \right|_{\varepsilon^\alpha}. \quad (29)$$

The *electrochemical potential*, incorporates the chemical and electrical energy required to insert the particular species into the mixture. In this formulation, this is

$$\tilde{\mu}^{\alpha_j} = \mu^{\alpha_j} + z^{\alpha_j} \left(\Lambda - \frac{\rho^\alpha}{\rho^{\alpha_j}} \left. \frac{\partial \tilde{A}^\alpha}{\partial z^{\alpha_j}} \right|_{\varepsilon^\alpha, \rho^{\alpha_j}, \dots} \right). \quad (30)$$

The additional terms are generically termed the electrical potential, and classically they are written as $\tilde{z}_{\alpha_j} e \psi$ where $\tilde{z}_{\alpha_j}$ is the charge number of α_j , e is the charge on an electron, and ψ is the electrical potential. The Lagrange multiplier, Λ , has units of energy/charge and is sometimes referred to as the streaming potential (Sasidhar and Ruckenstein, 1981). The first part of the electrical potential, $z^{\alpha_j} \Lambda$, has been derived before (Huyghe and Janssen, 1999; Gu *et al.*, 1999), and is the energy associated with inserting a charged ion into a system when the system wants to

remain charge neutral. The second term is novel, and represents the energy of changing the charge of a particular species if, for example, an electron were added to a molecule. Interchanging between the different independent variables z^{α_j} and $\tilde{z}_{\alpha_j}$, requires the molar weight, M^{α_j} [mass α_j per mole α_j], and Faraday's constant, F_c (96000 C/mole): $\partial \tilde{A}^\alpha / \partial z^{\alpha_j} = (M^{\alpha_j} / F_c)(\partial \tilde{A}^\alpha / \partial \tilde{z}_{\alpha_j})$. If the system is not charge neutral ($\Lambda = 0$) and the charge number of the species is constant, then the electrochemical potential reduces to the chemical potential, that is, the energy required to insert an atom in the presence of the electric field, $\mathbf{E}_T$, which enters into the chemical potential through the definition of $\tilde{A}^\alpha$ – see Equation (6).

With (29) as the definition of the chemical potential and by using (23) we can determine the relationship between the chemical potential and the partial stress tensors. By eliminating the Lagrange multipliers and using the definition of μ^{sj} , (23) becomes

$$\mathbf{t}^{sj} - \frac{\rho^{sj}}{\rho^{sN}} \mathbf{t}^{sN} = \left[-\rho^{sj}(\mu^{sj} - \mu^{sN}) + \rho^{sj}(\tilde{A}^{sj} - \tilde{A}^{sN}) + \rho^{sj}(z^{sj} - z^{sN})\Lambda \right] \mathbf{I}. \quad (31)$$

Summing these equation on j from 1 to N and using

$$\sum_{j=1}^N \rho^{sj} \mu^{sj} = \rho^s \tilde{A}^s + p^s, \quad (32)$$

$$\sum_{j=1}^N \mathbf{t}^{sj} = -(p^s - q_e^s \Lambda) \mathbf{I} + \mathbf{t}_e^s + \frac{\varepsilon^l}{\varepsilon^s} \mathbf{t}_s^l + \mathbf{t}_{as}^s, \quad (33)$$

where $\mathbf{t}_{as}^s$ indicates the anti-symmetric portion of the stress tensor, $\mathbf{t}_{as}^s = \mathbf{t}^s - \mathbf{t}_{sym}^s$, one obtains an expression for μ^{sN} , which when substituted back into Equation (31) yields

$$\mu^{sj} \mathbf{I} = \tilde{A}^{sj} \mathbf{I} - \frac{1}{\rho^{sj}} \mathbf{t}^{sj} + \frac{1}{\rho^s} \left(\mathbf{t}_e^s + \frac{\varepsilon^l}{\varepsilon^s} \mathbf{t}_s^l + \mathbf{t}_{as}^s \right) + z^{sj} \Lambda \mathbf{I}, \quad j = 1, \dots, N. \quad (34)$$

The first two terms on the right-hand-side form the Bowen chemical potential (see Bowen, 1976). In Bennethum *et al.* (1996, 2000) it was shown that incorporating the stress tensors yields an appropriate definition of chemical potential when no electric field or charges exist. Note that changing the effective stress or hydration stress in a porous medium results in a change in the chemical potential, which in turn produces a different balance of species within phases. The last term incorporates charge neutrality and suggests that changing the location of charges changes the *chemical* potential as well.

Taking advantage of these relations and simplifying gives us the dissipative portion of the entropy inequality in Appendix A.

4. Near-Equilibrium Constitutive Restrictions

Equilibrium is defined to occur when the following variables, defined generically as x_a , are zero:

$$\mathbf{d}^l, \dot{\varepsilon}^l, \mathbf{v}^{l,s}, \nabla \mathbf{v}^{lj,l}, \mathbf{v}^{\alpha_j,\alpha}, \quad \alpha = l, s \quad j = 1, \dots, N-1 \quad (35)$$

$$\varepsilon^l \rho^{lj} \hat{\mathbf{e}}_s^{lj}, \boldsymbol{\omega}^\alpha, \varepsilon^l \rho^{lj} \hat{\mathbf{Z}}_s^{lj}, \quad \alpha = l, s, \quad j = 1, \dots, N. \quad (36)$$

Using a dimensionality argument we can show that these variables are functionally independent. Hence, we have

$$\left. \frac{\partial}{\partial x_a} \left(\sum_{\alpha=l,s} \varepsilon^\alpha \rho^\alpha T \hat{\Lambda}^\alpha \right) \right|_{D|_e} = 0, \quad \left. \frac{\partial^2}{\partial (x_a x_b)} \left(\sum_{\alpha=l,s} \varepsilon^\alpha \rho^\alpha T \hat{\Lambda}^\alpha \right) \right|_{D|_e} \geq 0, \quad (37)$$

where subscript e denotes equilibrium. Note that we have not incorporated $\hat{\mathbf{r}}^{\alpha_j}$ and $\hat{\mathbf{q}}^{\alpha_j}$ into the above set. This is because without incorporating specific chemical reactions, incorrect results are obtained.

So for example, considering $x_a = \mathbf{d}^l$, we have at equilibrium

$$\mathbf{t}_{\text{sym}}^l = -(p^l - q_e^l \Lambda) \mathbf{I}. \quad (38)$$

To obtain results which hold near equilibrium, we expand linearly about equilibrium. For example

$$\begin{aligned} & \left(\mathbf{t}_{\text{sym}}^l + (p^l - q_e^l \Lambda) \mathbf{I} + \sum_{j=1}^N \rho^{lj} \mathbf{v}^{lj,l} \mathbf{v}^{lj,l} \right)_{\text{neq}} \\ & \approx \left(\mathbf{t}_{\text{sym}}^l + (p^l - q_e^l \Lambda) \mathbf{I} \right)_{\text{eq}} + \mathbf{f}_1 : \mathbf{d}^l + f_2 \dot{\varepsilon}^l + \dots, \end{aligned} \quad (39)$$

where $\mathbf{f}_1$ is a fourth order tensor and f_2 is a scalar. These material coefficients are functions of all independent variables which are not in the lists (35) or (36). In this manner one can obtain cross effects, for example, Eringen (1998), and nonlinear terms, for example, Bennethum and Giorgi (1997). With the exception of $\mathbf{v}^{l,s}$ and $\mathbf{v}^{\alpha_j,\alpha}$, we choose to linearize only about the one variable which produces a quadratic term in the entropy inequality, for example for the liquid phase stress tensor

$$\mathbf{t}_{\text{sym}}^l \approx -(p^l - q_e^l \Lambda) \mathbf{I} + \mathbf{v} : \mathbf{d}^l - \sum_{j=1}^N \rho^{lj} \mathbf{v}^{lj,l} \mathbf{v}^{lj,l}, \quad (40)$$

where $\mathbf{v}$ is a fourth-order tensor. Note that similar to (28), it is the term $p^l - q_e^l \Lambda$ which represents normal force per unit area.

We remark here that in order to obtain equilibrium results the variables listed in (35) and (36) must be functionally independent. However, near equilibrium, one can linearize about any independent variable which is zero at equilibrium.

We define the swelling pressure as

$$\pi^\alpha = -\varepsilon^\alpha \rho^\alpha \left. \frac{\partial \tilde{A}^l}{\partial \varepsilon^\alpha} \right|_{\rho^{\alpha j}, T, \dots}, \quad (41)$$

which represents the change in energy of phase α with respect to the relative quantity of phase α in the system. If the liquid and solid phase are non-interacting (non-swelling), then the energy of either phase would not change if the volume fractions were changed (keeping the densities fixed) and this quantity would be zero. A more detailed discussion of p^α and π^α is forthcoming in another manuscript.

Using these definitions and linearizing about $\hat{\varepsilon}^l$, we obtain

$$\mu^l \hat{\varepsilon}^l = p^l + \pi^l - (p^s + \pi^s) + \frac{1}{2} \varepsilon_o \mathbf{E}_T \cdot (\mathbf{E}^l - \mathbf{E}^s), \quad (42)$$

where μ^l is the material coefficient and is not to be confused with the chemical potential. Thus, if there are no effects of the electric field and $p^l + \pi^l > p^s + \pi^s$ the volume fraction will change so as to increase the amount of the liquid phase. The last term involves the differences in a portion of the electrostress tensor $\mathbf{t}_E = \mathbf{D}\mathbf{E} - \frac{1}{2} \varepsilon_o \mathbf{E} \cdot \mathbf{E}\mathbf{I}$ (see the conservation of linear momentum equation in Part I). Thus, if the contribution of the electric field of one phase is greater than the other, then the equilibrium volume fraction will be affected, but the electrical potential does *not* effect the equilibrium volume fraction.

Linearizing about $\boldsymbol{\omega}^\alpha = \nabla \mathbf{v}^l - \mathbf{d}^l$ yields the traditional result

$$\varepsilon^\alpha \mathbf{t}_{as}^\alpha = \mathbf{Q}^\alpha : \boldsymbol{\omega}^\alpha, \quad (43)$$

where $\mathbf{Q}^\alpha$ is a second-order tensor which may be a function of all independent variables not necessarily zero at equilibrium, $\boldsymbol{\omega}^\alpha$ is the anti-symmetric part of the gradient of the velocity of phase α , and $\mathbf{t}_{as}^\alpha$ is the anti-symmetric part of the stress tensor. This expression states that the stress tensors are in general not symmetric, which was known from the conservation of angular momentum (see Part I (Bennethum and Cushman, 2002)). Cross effects can also be obtained (Eringen, 1998).

Linearizing about the conservation of charge exchange term, $\hat{Z}_s^{lj}$, yields the following nearequilibrium result:

$$\varepsilon^l \rho^{lj} G^{lj} \hat{Z}_s^{lj} = \frac{\rho^s}{\rho^{sj}} \frac{\partial \tilde{A}^s}{\partial z^{sj}} - \frac{\rho^l}{\rho^{lj}} \frac{\partial \tilde{A}^l}{\partial z^{lj}}, \quad (44)$$

where G^{lj} is the linearization constant and where we assumed there are no cross effects. Thus, a transfer of ions of one species between phases occurs only if there is an imbalance in the part of the electric-potential involving ion disassociation.

Adsorption relations are obtained by linearizing about the rate at which mass is transferred from the solid phase to the liquid phase, $\varepsilon^l \rho^{lj} \hat{e}_s^{lj}$. At equilibrium we obtain

$$\mu^{lj} - \mu^{sj} = (z^{sj} - z^{lj}) \Lambda + \left(\frac{z^{lj} \rho^l}{\rho^{lj}} \frac{\partial \tilde{A}^l}{\partial z^{lj}} - \frac{z^{sj} \rho^s}{\rho^{sj}} \frac{\partial \tilde{A}^s}{\partial z^{sj}} \right), \quad (45)$$

or

$$\tilde{\mu}^{lj} = \tilde{\mu}^{sj}, \quad (46)$$

which is why we defined the electrochemical potential as we did. This is the boundary condition between phases, and in a more general framework (see e.g. Bennethum and Cushman (1996a, b)) this would produce the boundary condition for an osmotic experiment in which the mixture on one side contains ions. If the medium is not charge neutral, then $\Lambda = 0$ and assuming the charge per species remains unchanged, the boundary condition becomes a balance of the chemical potential.

To obtain a nearequilibrium result governing phase transition, we linearize about $\hat{e}_\beta^{lj}$. Neglecting quadratic terms of relative velocities yields

$$\rho^{lj} \mathbf{K}^{lj} \hat{e}_s^{lj} = \tilde{\mu}^{sj} - \tilde{\mu}^{lj} + \rho^{lj} z^{sj} G^{lj} \hat{\mathbf{Z}}_s^{lj}, \quad (47)$$

where, by (44), the last term on the right-hand-side is zero at equilibrium. Thus the larger the difference in the electrochemical potentials, the faster the phase transitions occur.

The relationship between the chemical potential and the partial stress tensor of the liquid phase is obtained exactly as in the previous section for the solid phase, except that the liquid phase result holds only at equilibrium

$$\mu^{lj} \mathbf{I} = \tilde{A}^{lj} \mathbf{I} - \frac{1}{\rho^{lj}} \mathbf{t}^{lj} + z^{lj} \Lambda \mathbf{I}. \quad (48)$$

5. BulkPhase Flow and Diffusion

The equations which govern momentum balance in porous media are known as generalized Darcy's equations, after Darcy, who in 1856 empirically derived the rather simple relationship that flux is proportional to the gradient in fluid pressure

$$\varepsilon^l \mathbf{v}^{l,s} = -\mathbf{K} \cdot (\nabla p^l + \varepsilon^l \rho^l \mathbf{g}), \quad (49)$$

where $\mathbf{K}$ is the conductivity of the material. It is generally thought to be valid for slow-moving viscous fluids through a homogeneous granular media. We would like to determine the generalization of this law for the swelling charged porous media considered here.

To begin with, we obtain a near equilibrium expression for $\hat{\mathbf{T}}_s^l$ by linearizing the coefficient of $\mathbf{v}^{l,s}$ about $\mathbf{v}^{l,s}$ and the diffusive velocities, $\mathbf{v}^{lj,l}$ in order to capture the effects of ion hydration (see e.g. (39)). This expression may then be substituted into the conservation of momentum equation. Neglecting inertial effects, the Brinkman correction term, $\nabla \cdot \nabla \mathbf{v}^l$, as well as the term involving $\nabla \cdot \boldsymbol{\omega}^l$ (see (43)), and using

(38) to eliminate $\mathbf{t}^l$ we obtain

$$\begin{aligned} \mathbf{R} \cdot \mathbf{v}^{l,s} = & -\varepsilon^l \nabla p^l + \varepsilon^l \rho^l (\mathbf{g}^l + \mathbf{g}_I) + \pi^l \nabla \varepsilon^l + \varepsilon^l q_e^l \mathbf{E}_T + \\ & + \varepsilon^l q_e^l \nabla \Lambda - \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \mathcal{E}^s} : (\nabla \mathcal{E}^s)^T - \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \mathbf{E}_T} \cdot (\nabla \mathbf{E}_T)^T - \\ & - \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial (\nabla T)} \cdot (\nabla^2 T) - \sum_{j=1}^N \mathbf{r}^{lj} \cdot \mathbf{v}^{lj,l}, \end{aligned} \quad (50)$$

where $\partial A^l / \partial \mathcal{E}^s : (\nabla \mathcal{E}^s)^T = (\partial A^l) / (\partial \mathcal{E}_{ij}^s) \mathcal{E}_{ij,k}^s$ in indicial notation. The material coefficients $\mathbf{R}$ and $\mathbf{r}^{lj}$ are second-order tensors which may be a function of all independent variables not equal to zero at equilibrium, including ε^l . The first two terms on the right-hand side recover the standard Darcy's Equation (49), where p^l is defined in (38). The third term involving $\mathbf{g}_I$, is due to fluctuations in the electric field, see its definition in Appendix A of Part I (Bennethum and Cushman, 2002). The terms not involving the electric field have been derived before (Bennethum and Cushman, 1996b) and these results are discussed in detail in Bennethum *et al.* (1997). They indicate that flow can be driven by a gradient in the volume fraction if the medium swells ($\pi^l \neq 0$) and also by gradients in shear strain, the latter of which may be appropriate for swelling media with low water content. These terms account for the chemical/hydration forces between the solid and liquid phase. The term involving $(\nabla^2 T)$ is likely negligible – first order thermal effects are obtained by considering cross terms. In addition to these terms, we have the Lorentz force ($\varepsilon^l q_e^l \mathbf{E}_T$), the Kelvin force (using Equation (21)) $\varepsilon^l \mathbf{P}^l \cdot \nabla \mathbf{E}_T$, and a term enforcing charge neutrality, $\varepsilon^l q_e^l \nabla \Lambda$. The last term involving summation over species is a cross-term put in specifically to capture the hydrating effect of ions. Each charged particle surrounds itself with water molecules, thereby impeding or enhancing bulk phase flow when the diffusive velocity of the charged species are non-negligible. This term is negligible for non-hydrating species.

Alternatively we can rewrite Darcy's law in terms of the bulk phase chemical potential, as this is the formulation often used in applications (Gu *et al.*, 1999; Huyghe and Janssen, 1999). Define the bulk phase potential as the Gibb's potential

$$G^l = \sum_{j=1}^N C^{lj} \mu^{lj} = \tilde{A}^l + \sum_{j=1}^N \rho^{lj} \frac{\partial \tilde{A}^l}{\partial \rho^{lj}} = \tilde{A}^l + \frac{p^l}{\rho^l}. \quad (51)$$

Further we wish to rewrite the electrical forcing terms in terms of the Maxwell stress tensor

$$\mathbf{t}_M^l = \mathbf{D}^l \mathbf{E}_T - \frac{1}{2} \varepsilon_o \mathbf{E}^l \cdot \mathbf{E}_T \mathbf{I}. \quad (52)$$

Assuming that the electric fields are defined such that the exchange term in Gauss law is zero ($\nabla \cdot (\varepsilon^l \mathbf{D}^l) - \varepsilon^l q_e^l = 0$) we have

$$\begin{aligned}
& \varepsilon^l q_e^l \mathbf{E}_T + \varepsilon^l \mathbf{P}^l \cdot \nabla \mathbf{E}_T \\
& = \nabla \cdot (\varepsilon^l \mathbf{D}^l \mathbf{E}_T - \frac{1}{2} \varepsilon_o \mathbf{E}^l \cdot \mathbf{E}_T \mathbf{I}) + \\
& \quad + \frac{1}{2} \varepsilon_o \mathbf{E}^l \cdot \mathbf{E}_T \nabla \varepsilon^l - \frac{1}{2} \varepsilon_o \varepsilon^l (\mathbf{E}^l \cdot \nabla \mathbf{E}_T - \nabla \mathbf{E}^l \cdot \mathbf{E}_T).
\end{aligned} \tag{53}$$

Assuming $\varepsilon^l \rho^l \mathbf{g}_I^l - \frac{1}{2} \varepsilon_o \varepsilon^l [\mathbf{E}^l \cdot \nabla \mathbf{E}_T - (\nabla \mathbf{E}^l) \cdot \mathbf{E}_T] = \sum_{j=1}^N \frac{1}{2} \varepsilon_o \varepsilon^l [< \nabla \mathbf{E}^j \cdot \mathbf{E} >^l - < \nabla \mathbf{E} \cdot \mathbf{E}^j >^l]$ (see Part I, Appendices B and C), is negligible, we have

$$\begin{aligned}
\mathbf{R} \cdot \mathbf{v}^{l,s} & = -\varepsilon^l \rho^l \nabla G^l + \sum_{j=1}^N \varepsilon^l (\rho^l)^2 \frac{\partial \tilde{A}^l}{\partial \rho^{lj}} \nabla C^{lj} + \varepsilon^l q_e^l \nabla \Lambda + \\
& \quad + \sum_{j=1}^N \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial z^{lj}} \nabla z^{lj} - \varepsilon^l \nabla \mathbf{E}_T \cdot \mathbf{P}^l + \varepsilon^l \rho^l \mathbf{g}^l + \nabla \cdot (\varepsilon^l \mathbf{t}_M^l) + \\
& \quad + \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial T} \nabla T + \frac{1}{2} \varepsilon_o \mathbf{E}_T \cdot \mathbf{E}^l \nabla \varepsilon^l - \sum_{j=1}^N \mathbf{r}^{lj} \cdot \mathbf{v}^{lj,l},
\end{aligned} \tag{54}$$

which is comparable to what is derived in Gu *et al.* (1999) and Huyghe and Janssen (1999). The first term is denoted as the *mechanochemical force* since it does not incorporate the electrical potential, the second term is due to osmotic effects, the third due to enforcing charge-neutrality, and the fourth term is due to changes in charges per species. The next two terms on the second line account for the Kelvin force and gravity, respectively, and the following term states that the Maxwell stress tensor effects flow. Thermal effects appear explicitly in this form, and since $\partial \tilde{A} / \partial T < 0$ (see (20)), we see that flow goes from hot to cold regions. The last term on the second line magnifies the effects due to gradients in volume fractions if electric fields are non-negligible. The last term is due to hydration and was discussed above.

Diffusion in a single-phase mixture is governed by *Fick's law*, which states diffusive velocity is proportional to the gradient of the chemical potential. Here we derive a novel form of Fick's law. Begin with the coefficient of $\mathbf{v}^{lj,l}$ in the residual entropy inequality which, when set to zero, gives at equilibrium:

$$\begin{aligned}
& \varepsilon^\alpha \rho^{\alpha_j} (\hat{\mathbf{i}}^{\alpha_j} + \hat{\mathbf{T}}_\beta^{\alpha_j}) - \varepsilon^\alpha \rho^{\alpha_j} (\hat{\mathbf{i}}^{\alpha_N} + \hat{\mathbf{T}}_\beta^{\alpha_N}) \\
& = -\nabla [\varepsilon^\alpha \rho^{\alpha_j} (\tilde{A}^{\alpha_j} - \tilde{A}^{\alpha_N})] - \\
& \quad - \varepsilon^\alpha \nabla \mathbf{E}_T \cdot (\mathbf{P}^{\alpha_j} - \frac{\rho^{\alpha_j}}{\rho^{\alpha_N}} \mathbf{P}^{\alpha_N}) + \frac{1}{2} \varepsilon_o \mathbf{E}_T \cdot (\mathbf{E}^{\alpha_j} - \frac{\rho^{\alpha_j}}{\rho^{\alpha_N}} \mathbf{E}^{\alpha_N}) \nabla \varepsilon^\alpha + \\
& \quad + \rho^\alpha \left(\frac{\partial \tilde{A}^\alpha}{\partial \rho^{\alpha_j}} - \frac{\partial \tilde{A}^\alpha}{\partial \rho^{\alpha_N}} \right) \nabla (\varepsilon^\alpha \rho^{\alpha_j}) + \\
& \quad + \varepsilon^\alpha \rho^\alpha \left(\frac{\partial \tilde{A}^\alpha}{\partial z^{\alpha_j}} \nabla z^{\alpha_j} - \frac{\rho^{\alpha_j}}{\rho^{\alpha_N}} \frac{\partial \tilde{A}^\alpha}{\partial z^{\alpha_N}} \nabla z^{\alpha_N} \right) -
\end{aligned}$$

$$\begin{aligned}
& -\Lambda \nabla (\varepsilon^\alpha \rho^{\alpha_j} (z^{\alpha_j} - z^{\alpha_N})) - \varepsilon^\alpha (q_e^{\alpha_j} - \frac{\rho^{\alpha_j}}{\rho^{\alpha_N}} q_e^{\alpha_N}) \mathbf{E}_T - \\
& - \varepsilon^\alpha \nabla \left(\frac{\rho^{\alpha_j}}{\rho^{\alpha_N}} \right) \cdot \mathbf{t}^{\alpha_N}, \quad j = 1, \dots, N-1.
\end{aligned} \tag{55}$$

Summing (55) over j from 1 to N and making use of the equilibrium relationship

$$\begin{aligned}
\sum_{j=1}^N \varepsilon^l \rho^{lj} \hat{\mathbf{T}}_s^{lj} &= \varepsilon^l \rho^l \hat{\mathbf{T}}_s^l \\
&= -\varepsilon^l \rho^l \nabla \tilde{A}^l + \sum_{j=1}^N \rho^l \frac{\partial \tilde{A}^l}{\partial \rho^{lj}} \nabla (\varepsilon^l \rho^{lj}) + \sum_{j=1}^N \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial z^{lj}} \nabla z^{lj} + \\
&\quad + \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial T} \nabla T - \varepsilon^l \nabla \mathbf{E}_T \cdot \mathbf{P}^l + \frac{1}{2} \varepsilon_o \mathbf{E}_T \cdot \mathbf{E}^l \nabla \varepsilon^l - \Lambda \nabla (\varepsilon^l q_e^l)
\end{aligned} \tag{56}$$

we obtain (57) for the case $j = N$. Substituting this result back into (55) and again making use of (29) and (48) we obtain the equilibrium result

$$\begin{aligned}
\varepsilon^l \rho^{lj} (\hat{\mathbf{i}}^{lj} + \hat{\mathbf{T}}_s^{lj}) &= -\varepsilon^l \rho^{lj} \nabla \mu^{lj} - \nabla \cdot (\varepsilon^l \mathbf{t}^{lj}) + \varepsilon^l q_e^{lj} \nabla \Lambda + \varepsilon^l \rho^{lj} \frac{\partial \tilde{A}^l}{\partial T} \nabla T - \\
&\quad - \varepsilon^l \nabla \mathbf{E}_T \cdot \mathbf{P}^{lj} + \varepsilon^l \rho^{lj} (z^l - z^{lj}) \mathbf{E}_T + \frac{1}{2} \varepsilon_o \mathbf{E}_T \cdot \mathbf{E}^{lj} \nabla \varepsilon^l.
\end{aligned} \tag{57}$$

Next we linearize the coefficient of $\mathbf{v}^{lj,l}$ about equilibrium by expanding the coefficient in the original entropy inequality given in Appendix B in terms of both $\mathbf{v}^{lj,l}$ and $\mathbf{v}^{l,s}$ so that Onsager's principal is still satisfied. This allows us to obtain nonrelative results since $j = 1, \dots, N$. Here we make use of the fact that $\tilde{A}^l$ and the primary independent variables, listed in (8), are the same at equilibrium and nearequilibrium. That this holds for $\tilde{A}^l$ is justified by $\tilde{A}^l$ not being a function of any of the variables which define equilibrium. Using this result to eliminate $\varepsilon^l \rho^{lj} (\hat{\mathbf{i}}^{lj} + \hat{\mathbf{T}}_s^{lj})$ in the conservation of momentum equation, neglecting the inertial term, and approximating μ^{lj} using Equation (48), we obtain

$$\begin{aligned}
\sum_{k=1}^N \mathbf{r}^{lj,k} \cdot \mathbf{v}^{lk,l} &= -\varepsilon^l \rho^{lj} \nabla \mu^{lj} + \varepsilon^l \rho^{lj} (\mathbf{g}^{lj} - \mathbf{g}_I^{lj}) + \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial z^{lj}} \nabla z^{lj} + \\
&\quad + \varepsilon^l q_e^{lj} \nabla \Lambda + \varepsilon^l \rho^{lj} z^l \mathbf{E}_T + \varepsilon^l \rho^{lj} \frac{\partial \tilde{A}^l}{\partial T} \nabla T - (\mathbf{r}^{lj})^T \cdot \mathbf{v}^{l,s},
\end{aligned} \tag{58}$$

where $\mathbf{r}^{lj,k}$ and $\mathbf{r}^{lj}$ are material coefficient tensors which satisfy the Onsager relationship. Note that this is similar in form to the bulk flow Equation (54). The Lorentz term, $\varepsilon^l \rho^{lj} z^l \mathbf{E}_T$, is actually a relative Lorentz term, and the form is a consequence of the bulk phase velocity, $\mathbf{v}^l$, being a mass-averaged velocity.

To analyze this further, assume that dependence on other diffusive velocities is negligible ($\mathbf{r}^{l_{jk}} = \mathbf{0}$ for $j \neq k$), the body force, $\mathbf{g}^{l_j}$ is gravity, $\mathbf{g}$, temperature gradients are negligible, the body force due to electric field fluctuations is negligible $\mathbf{g}_I^{l_j} = 0$, and charge neutrality is enforced with the Lagrange multiplier, Λ . In this case (58) simplifies to

$$\begin{aligned} \mathbf{r}^{l_{jj}} \cdot \mathbf{v}^{l_{j,l}} = & -\varepsilon^l \rho^{l_j} \nabla \tilde{\mu}^{l_j} - \varepsilon^l \rho^{l_j} \Lambda \nabla z^{l_j} + \varepsilon^l \rho^{l_j} \mathbf{g} - \\ & -\varepsilon^l C^{l_j} q_e^l \mathbf{E}_T - (\mathbf{r}^{l_j})^T \cdot \mathbf{v}^{l,s}. \end{aligned} \quad (59)$$

The first two terms (on the right-hand side) state that the driving forces are the electro-chemical potential and fluctuations in z^{l_j} . The Lorentz term involves the bulk-phase force weighted by the mass fraction, just as the gravitational force is weighted by the mass density. This term is new and should be evaluated carefully, although similar bulk terms have been derived before (Benach and Müller, 1973).

In this form, we can easily compare the results with those obtained in Gu *et al.* (1999) and Huyghe and Janssen (1999). Citing directly from Gu *et al.* (Eqs. (35) and (9) in 1999) we have

$$d_\alpha \mathbf{v}^{\alpha_j} = -\rho^\alpha \nabla \tilde{\mu}^\alpha + \sum_j f_{ij} \mathbf{v}^{\alpha_j} \quad \text{where} \quad \tilde{\mu}^{\alpha_j} = \frac{\partial W}{\partial \rho^{\alpha_j}} + \frac{p^\alpha}{\rho_T^{\alpha_j}} + \frac{F_c \Psi \tilde{z}_{\alpha_j}}{M^{\alpha_j}},$$

where the notation has been slightly modified to correspond more closely with that used in this paper. Here W is the inner part of the Helmholtz energy density of total mixture (per unit of undeformed mixture volume), ρ^{α_j} is the mass density per unit of current mixture volume, $\rho_T^{\alpha_j}$ is the true mass density of α_j , F_c is Faraday's constant, Ψ is the electric potential, $\tilde{z}_{\alpha_j}$ is the valence of α_j with sign, and M^{α_j} is the molar weight of species α_j .

In Huyghe and Janssen (Eqs. (50) and (39) in 1999), the results are presented in the Lagrangian framework and are given as

$$\begin{aligned} \sum_k \mathbf{B}^{jk} \cdot \mathbf{v}^{l_{k,s}} + \mathbf{Q}^{l_j} \cdot \mathbf{q}^{l_{js}} = & -\varepsilon^{l_j} \nabla_o \tilde{\mu}^{l_j} + \mathbf{F}^T \cdot \varepsilon^l \rho^{l_j} \mathbf{g}, \\ \text{where} \quad \tilde{\mu}^{l_j} = & \frac{\partial W}{\partial N^{l_j}} + p^l + \frac{\tilde{z}_{l_j} \lambda}{V^{l_j}}, \end{aligned}$$

where we have again modified the notation slightly. Here $\mathbf{q}^{l_{js}}$ is the Lagrangian heat flux, $\mathbf{F}^T$ is the transpose of the deformation gradient of the solid phase, λ is the electrical potential of the medium multiplied by the constant of Faraday, W is Helmholtz free energy per unit initial mixture volume, and V^{l_j} is the molar volume of constituent l_j .

Comparing the results, we see that if z^{l_j} in (59) is constant and if there is no electric field, then the results correspond up to the units which the terms are defined. Recall that $\tilde{z}_{\alpha_j} = M^{\alpha_j} z^{\alpha_j} / F_c$ (see paragraph following (30)), so that the term involving the electric potentials all correspond. The presence of the pressure

in the definitions of the electro-chemical potentials are a result of the above two equations being an expression for the absolute velocity and velocity relative to the solid phase, respectively, instead of the diffusive velocity relative to the barycentric liquid phase velocity. The presence of the Lagrangian heat flux vector in Huyghe and Janssen's formulation is a result of further linearization to get cross-effects between the heat flux and the velocities.

The last term in all three versions of (59) is the result of linearizing the dissipative terms about both $\mathbf{v}^{l,s}$ and $\mathbf{v}^{lj,l}$ and accounts for hydrating effects.

6. Discussion

We exploited the entropy inequality to obtain restrictions on the form of constitutive relations for swelling porous media composed of a possibly polarizable solid and liquid phase, with charges and an electric field. This has applications in swelling clay soils, biopolymers, biological membranes, pulsed electrophoresis, chromatography, drug delivery, and other swelling systems. We did so under the philosophy that it is the total electric field which contributes to the force and work terms in the conservation of momentum and energy, and that it is only the total electric field which is measurable. This produced an additional forcing term involving the gradient of the volume fraction, and an additional body force which is a result of fluctuations in the electric field, appearing in the macroscale conservation of momentum equation. The new body force term appears wherever gravity appears, and the extra term involving the gradient of the volume fraction, $\frac{1}{2}\varepsilon_o \mathbf{E}_T \cdot \mathbf{E}^l \nabla \varepsilon^l$, affects only the bulk phase flow when this equation is written in terms of the Maxwell stress tensor of the liquid phase, see (54). These terms are a consequence of assuming that the primitive form of the momentum and energy equation is the form written in terms of the Lorentz force and Kelvin force, instead of the Maxwell stress tensor (see Part I).

The Lagrange multiplier which enforces charge neutrality is shown to correspond with an electric potential as is done classically (Hill, 1968; Gu *et al.*, 1999; Huyghe and Janssen, 1999). Care must be taken in applying charge neutrality, as it often does not hold. In the case where it does not hold, Λ must be set to zero, and the resulting equations still apply. In the case where charge neutrality does apply, it is seen to affect the macroscopic stress tensors (see e.g. (28) and (40)) so that the trace of the liquid stress tensor is not proportional to the thermodynamically defined pressure, p^α , but is proportional to $p^\alpha - q_e^\alpha \Lambda$. Thus care must be taken in interpreting the pressure term given in the generalized Darcy law, (50). When charge neutrality does not apply and the charge per species does not change, the electrochemical potential reduces to the thermodynamically defined chemical potential where the modified Helmholtz potential is a function of the total electric field and charges.

It was assumed that ion-disassociation can occur, and this is represented by terms which contain the partial of the Helmholtz free energy density with respect to

the charge per unit mass. This term can also be related to the charge per molecule, see the discussion following (30). This term does not affect pressure, but does manifest itself in bulk phase flow and diffusion.

The boundary condition between phases is a natural by-product of this formulation, and it is shown that the electric field does not influence the boundary condition, but that it is the balance of the electrochemical potential which determines equilibrium, see (46). Polarization enters into it only through the definition of $\tilde{A}^\alpha$, see (6).

It was also shown that the rate at which the medium swells is determined by the difference in the thermodynamic and swelling pressures, $(p^l + \pi^l) - (p^s + \pi^s)$ and the difference of a portion of the electro-stress tensor, see Equation (42).

Appendix A. Dissipative Entropy Inequality

The dissipative portion of the entropy inequality is

$$\begin{aligned}
 & \left(\sum_{\alpha} \varepsilon^{\alpha} \rho^{\alpha} T \hat{\Lambda}^{\alpha} \right)_D \\
 &= \mathbf{d}^l : \left[\varepsilon^l \mathbf{t}_{\text{sym}}^l + \sum_{j=1}^N \varepsilon^l \rho^{lj} \mathbf{v}^{lj,l} \mathbf{v}^{lj,l} + \varepsilon^l p^l - \varepsilon^l q_e^l \Lambda \right] + \\
 &+ \dot{\varepsilon}^l \left[-\varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \varepsilon^l} - \varepsilon^s \rho^s \frac{\partial \tilde{A}^s}{\partial \varepsilon^l} + p^l - p^s + \frac{1}{2} \varepsilon_o \mathbf{E}_T \cdot (\mathbf{E}^l - \mathbf{E}^s) \right] + \\
 &+ \mathbf{v}^{l,s} \cdot \left[-\varepsilon^l \rho^l \nabla \tilde{A}^l + \sum_{j=1}^N \rho^l \frac{\partial \tilde{A}^l}{\partial \rho^{lj}} \nabla (\varepsilon^l \rho^{lj}) + \sum_{j=1}^N \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial z^{lj}} \nabla z^{lj} + \right. \\
 &+ \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial T} \nabla T - \varepsilon^l \nabla \mathbf{E}_T \cdot \mathbf{P}^l + \\
 &+ \left. \frac{1}{2} \varepsilon_o \mathbf{E}_T \cdot \mathbf{E}^l \nabla \varepsilon^l - \Lambda \nabla (\varepsilon^l q_e^l) - \varepsilon^l \rho^l \hat{\mathbf{T}}_s^l \right] + \\
 &+ \sum_{j=1}^{N-1} \nabla \mathbf{v}^{lj,l} : \varepsilon^l \left[\mathbf{t}^{lj} - \frac{\rho^{lj}}{\rho^{lN}} \mathbf{t}^{lN} + \rho^l \rho^{lj} \left(\frac{\partial \tilde{A}^l}{\partial \rho^{lj}} - \frac{\partial \tilde{A}^l}{\partial \rho^{lN}} \right) - \right. \\
 &- \rho^{lj} (\tilde{A}^{lj} - \tilde{A}^{lN}) \mathbf{I} - (q_e^{lj} - \frac{\rho^{lj}}{\rho^{lN}} q_e^{lN}) \Lambda \mathbf{I} \left. \right] + \\
 &+ \sum_{\alpha=l,s} \sum_{j=1}^{N-1} \mathbf{v}^{\alpha_j, \alpha} \cdot \left[-\varepsilon^{\alpha} \rho^{\alpha_j} (\hat{\mathbf{i}}^{\alpha_j} + \hat{\mathbf{T}}_{\beta}^{\alpha_j}) + \varepsilon^{\alpha} \rho^{\alpha_j} (\hat{\mathbf{i}}^{\alpha_N} + \hat{\mathbf{T}}_{\beta}^{\alpha_N}) - \right. \\
 &- \nabla (\varepsilon^{\alpha} \rho^{\alpha_j} (A^{\alpha_j} - A^{\alpha_N})) - \varepsilon^{\alpha} \nabla \mathbf{E}_T \cdot \left(\mathbf{P}^{\alpha_j} - \frac{\rho^{\alpha_j}}{\rho^{\alpha_N}} \mathbf{P}^{\alpha_N} \right) + \left. \right]
 \end{aligned}$$

$$\begin{aligned}
& + \frac{1}{2} \varepsilon_o \mathbf{E}_T \cdot \left(\mathbf{E}^{\alpha_j} - \frac{\rho^{\alpha_j}}{\rho^{\alpha_N}} \mathbf{E}^{\alpha_N} \right) \nabla \varepsilon^\alpha + \rho^\alpha \left(\frac{\partial \tilde{A}^\alpha}{\partial \rho^{\alpha_j}} - \frac{\partial \tilde{A}^\alpha}{\partial \rho^{\alpha_N}} \right) \nabla (\varepsilon^\alpha \rho^{\alpha_j}) + \\
& + \varepsilon^\alpha \rho^\alpha \left(\frac{\partial \tilde{A}^\alpha}{\partial z^{\alpha_j}} \nabla z^{\alpha_j} - \frac{\rho^{\alpha_j}}{\rho^{\alpha_N}} \frac{\partial \tilde{A}^\alpha}{\partial z^{\alpha_N}} \nabla z^{\alpha_N} \right) - \\
& - \Lambda (\nabla (\varepsilon^\alpha \rho^{\alpha_j} z^{\alpha_j}) - \nabla (\varepsilon^\alpha \rho^{\alpha_j} z^{\alpha_N})) - \\
& - \varepsilon^\alpha \left(q_e^{\alpha_j} - \frac{\rho^{\alpha_j}}{\rho^{\alpha_N}} q_e^{\alpha_N} \right) \mathbf{E}_T - \varepsilon^\alpha \nabla \left(\frac{\rho^{\alpha_j}}{\rho^{\alpha_N}} \right) \cdot \mathbf{t}^{\alpha_N} \Big] + \\
& + \sum_{\alpha=l,s} \frac{\varepsilon^\alpha}{T} \nabla T \cdot \left[\mathbf{q}^\alpha + \sum_{j=1}^N \left(\rho^{\alpha_j} \mathbf{v}^{\alpha_j, \alpha} \left(\tilde{A}^{\alpha_j} + \frac{1}{2} (\mathbf{v}^{\alpha_j, \alpha})^2 \right) - \right. \right. \\
& \left. \left. - \mathbf{t}^{\alpha_j} \cdot \mathbf{v}^{\alpha_j, \alpha} \right) - \rho^\alpha T \left(\eta^\alpha + \frac{\partial \tilde{A}^\alpha}{\partial T} \right) \mathbf{v}^{\alpha, s} \right] + \\
& + \sum_{\alpha=l,s} \sum_{j=1}^N \varepsilon^l \rho^{l_j} \tilde{e}_s^{l_j} \left[-\tilde{A}^l + \tilde{A}^s - \rho^l \frac{\partial \tilde{A}^l}{\partial \rho^{l_j}} + \rho^s \frac{\partial \tilde{A}^s}{\partial \rho^{s_j}} + \right. \\
& + \frac{\rho^s}{\rho^{s_j}} \frac{\partial \tilde{A}^s}{\partial z^{s_j}} (z^{l_j} - z^{s_j}) + (z^{l_j} - z^{s_j}) \Lambda + \\
& + \frac{1}{2} \mathbf{v}^{s_j, s} \cdot \mathbf{v}^{s_j, s} - \frac{1}{2} \mathbf{v}^{l_j, l} \cdot \mathbf{v}^{l_j, l} - \frac{1}{2} \mathbf{v}^{l, s} \cdot \mathbf{v}^{l, s} \Big] + \\
& + \sum_{\alpha=l,s} \sum_{j=1}^N \varepsilon^\alpha \rho^{\alpha_j} \hat{r}^{\alpha_j} \left[-\rho^\alpha \frac{\partial \tilde{A}^\alpha}{\partial \rho^{\alpha_j}} + z^{\alpha_j} \Lambda - \frac{1}{2} (\mathbf{v}^{\alpha_j, \alpha})^2 \right] + \\
& + \sum_{\alpha=l,s} \boldsymbol{\omega}^\alpha : [\varepsilon^\alpha \mathbf{t}_{as}^\alpha] + \sum_{\alpha=l,s} \sum_{j=1}^N \nabla \cdot (\varepsilon^\alpha \mathcal{J}^{\alpha_j}) \left[\rho^\alpha \frac{\partial \tilde{A}^\alpha}{\partial q_e^{\alpha_j}} - \Lambda \right] + \\
& + \sum_{\alpha=l,s} \varepsilon^\alpha \mathcal{J}^\alpha \cdot \mathbf{E}_T - \sum_{\alpha=l,s} \sum_{j=1}^N \varepsilon^\alpha \hat{q}^{\alpha_j} \left[\rho^\alpha \frac{\partial \tilde{A}^\alpha}{\partial q_e^{\alpha_j}} - \Lambda \right] + \\
& + \sum_{j=1}^N \varepsilon^l \rho^{l_j} \hat{Z}_s^{l_j} \left[\frac{\rho^s}{\rho^{s_j}} \frac{\partial \tilde{A}^s}{\partial z^{s_j}} - \frac{\rho^l}{\rho^{l_j}} \frac{\partial \tilde{A}^l}{\partial z^{l_j}} \right] \geq 0, \tag{60}
\end{aligned}$$

where subscripted sym and as mean the symmetric and anti-symmetric part of the tensor, respectively.

Appendix B. Entropy Inequality

The entropy inequality in its entirety is

$$\begin{aligned}
& \sum_{\alpha=l,s} \varepsilon^\alpha \rho^\alpha T \hat{\Lambda}^\alpha \\
&= \sum_{\alpha=l,s} \sum_{j=1}^N \varepsilon^\alpha \frac{D^s \rho^{\alpha_j}}{Dt} \left[-\rho^\alpha \frac{\partial \tilde{A}^\alpha}{\partial \rho^{\alpha_j}} + \lambda_\rho^{\alpha_j} + \tilde{A}^{\alpha_j} + \frac{1}{\rho^{\alpha_j}} \mathbf{E}_T \cdot \mathbf{P}^{\alpha_j} + z^{\alpha_j} \Lambda \right] + \\
&+ \dot{\mathbf{E}}_T \left[-\varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \mathbf{E}_T} - \varepsilon^s \rho^s \frac{\partial \tilde{A}^s}{\partial \mathbf{E}_T} - \varepsilon^l \mathbf{P}^l - \varepsilon^s \mathbf{P}^s \right] + \\
&+ \dot{T} \left[-\varepsilon^l \rho^l \eta^l - \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial T} - \varepsilon^s \rho^s \eta^s - \varepsilon^s \rho^s \frac{\partial \tilde{A}^s}{\partial T} \right] + \\
&+ \mathbf{d}^l : \left[\varepsilon^l \mathbf{t}_{\text{sym}}^l + \varepsilon^l \mathbf{P}^l \cdot \mathbf{E}_T \mathbf{I} + \sum_{j=1}^N \varepsilon^l \rho^{l_j} (\lambda_\rho^{l_j} \mathbf{I} + \tilde{A}^{l_j} \mathbf{I} + \mathbf{v}^{l_j,l} \mathbf{v}^{l_j,l}) \right] + \\
&+ \mathbf{d}^s : \left[\varepsilon^s \mathbf{t}_{\text{sym}}^s + \varepsilon^s \mathbf{P}^s \cdot \mathbf{E}_T \mathbf{I} + \sum_{j=1}^N \varepsilon^s \rho^{s_j} (\lambda_\rho^{s_j} \mathbf{I} + \tilde{A}^{s_j} \mathbf{I} + \mathbf{v}^{s_j,s} \mathbf{v}^{s_j,s}) - \right. \\
&- \varepsilon^l \rho^l \nabla \mathbf{F}^s \cdot \frac{\partial \tilde{A}^l}{\partial \mathcal{E}^s} \cdot (\nabla \mathbf{F}^s)^T - \varepsilon^s \rho^s \nabla \mathbf{F}^s \cdot \frac{\partial \tilde{A}^s}{\partial \mathcal{E}^s} \cdot (\nabla \mathbf{F}^s)^T \left. \right] + \\
&+ \sum_{\alpha=l,s} \boldsymbol{\omega}^\alpha : [\varepsilon^\alpha \mathbf{t}_{\text{as}}^\alpha] + \sum_{\alpha=l,s} \sum_{j=1}^{N-1} \frac{D^s \mathbf{v}^{\alpha_j,\alpha}}{Dt} \cdot \left[-\varepsilon^\alpha \rho^\alpha \frac{\partial \tilde{A}^\alpha}{\partial \mathbf{v}^{\alpha_j,\alpha}} \right] + \\
&+ \nabla \dot{T} \cdot \left[-\varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \nabla T} - \varepsilon^s \rho^s \frac{\partial \tilde{A}^s}{\partial \nabla T} \right] + \\
&+ \sum_{\alpha=l,s} \sum_{j=1}^N \nabla \frac{D^s \rho^{\alpha_j}}{Dt} \cdot \left[-\varepsilon^\alpha \rho^\alpha \frac{\partial \tilde{A}^\alpha}{\partial (\nabla \rho^{\alpha_j})} \right] + \\
&+ \dot{\varepsilon}^l \left[\sum_{j=1}^N (\rho^{l_j} \lambda_\rho^{l_j} - \rho^{s_j} \lambda_\rho^{s_j}) + \rho^l A^l - \rho^s A^s + \frac{1}{2} \varepsilon_o \mathbf{E}_T \cdot (\mathbf{E}^l - \mathbf{E}^s) + \right. \\
&+ \mathbf{E}_T \cdot (\mathbf{P}^l - \mathbf{P}^s) + \Lambda (q_e^l - q_e^s) - \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \varepsilon^l} - \varepsilon^s \rho^s \frac{\partial \tilde{A}^s}{\partial \varepsilon^l} \left. \right] + \\
&+ \dot{\mathbf{v}}^{l,s} \cdot \left[-\varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \mathbf{v}^{l,s}} \right] + \dot{\mathbf{d}}^l : \left[-\varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \mathbf{d}^l} \right] + \dot{\mathbf{w}}^l : \left[-\varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \mathbf{w}^l} \right] + \\
&+ \nabla \dot{\mathbf{v}}^{l,l} : \left[-\varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial (\nabla \mathbf{v}^{l,l})} \right] + \mathbf{v}^{l,s} \cdot \left[-\varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \varepsilon^l} \nabla \varepsilon^l - \right. \\
&- \sum_{j=1}^N \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \rho^{l_j}} \nabla \rho^{l_j} - \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \mathcal{E}^s} : (\nabla \mathcal{E}^s)^T - \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \mathbf{E}_T} \cdot (\nabla \mathbf{E}_T)^T - \\
&- \sum_{j=1}^N \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial z^{l_j}} \nabla z^{l_j} - \varepsilon^l \nabla \mathbf{E}_T \cdot \mathbf{P}^l - \varepsilon^l \rho^l \hat{\mathbf{T}}_s^l + \frac{1}{2} \varepsilon_o \mathbf{E}_T \cdot \mathbf{E}^l \nabla \varepsilon^l +
\end{aligned}$$

$$\begin{aligned}
& + \sum_{j=1}^N \left(\tilde{A}^{l_j} + \frac{1}{\rho^{l_j}} \mathbf{E}_T \cdot \mathbf{P}^{l_j} \right) \nabla(\varepsilon^l \rho^{l_j}) + \sum_{j=1}^N \lambda_\rho^{l_j} \nabla(\varepsilon^l \rho^{l_j}) + \\
& + \sum_{j=1}^N \varepsilon^l \rho^{l_j} \lambda_{q_e}^{l_j} \nabla z^{l_j} - \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \mathbf{v}^{l,s}} \cdot (\nabla \mathbf{v}^{l,s})^T - \sum_{j=1}^{N-1} \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \mathbf{v}^{l_j,l}} \cdot (\nabla \mathbf{v}^{l_j,l})^T - \\
& - \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \nabla T} \cdot \nabla^2 T - \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \nabla \mathbf{v}^{l_j,l}} : \nabla^2 \mathbf{v}^{l_j,l} - \sum_{j=1}^N \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial (\nabla \rho^{l_j})} \cdot \nabla^2 \rho^{l_j} - \\
& - \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \mathbf{d}^l} : (\nabla \mathbf{d}^l)^T - \varepsilon^l \rho^l \frac{\partial \tilde{A}^l}{\partial \mathbf{w}^l} : (\nabla \mathbf{w}^l)^T \Big] + \\
& + \sum_{\alpha=l,s} \sum_{j=1}^N (\nabla \mathbf{v}^{\alpha_j,\alpha})^T : \left[\varepsilon^\alpha \mathbf{t}^{\alpha_j} + \varepsilon^\alpha \mathbf{P}^{\alpha_j} \cdot \mathbf{E}_T \mathbf{I} + \varepsilon^\alpha \rho^{\alpha_j} \lambda_\rho^{\alpha_j} \mathbf{I} \right] + \\
& + \sum_{\alpha=l,s} \frac{\varepsilon^\alpha}{T} \nabla T \cdot \left[\mathbf{q}^\alpha + \sum_{j=1}^N \left(\rho^{\alpha_j} \mathbf{v}^{\alpha_j,\alpha} \left(\tilde{A}^{\alpha_j} + \frac{1}{2} (\mathbf{v}^{\alpha_j,\alpha})^2 \right) - \right. \right. \\
& \left. \left. - \mathbf{t}^{\alpha_j} \cdot \mathbf{v}^{\alpha_j,\alpha} \right) - T \rho^\alpha \left(\eta^\alpha + \frac{\partial \tilde{A}^\alpha}{\partial T} \right) \mathbf{v}^{\alpha,s} \right] + \\
& + \sum_{\alpha=l,s} \sum_{j=1}^N \mathbf{v}^{\alpha_j,\alpha} \cdot \left[-\varepsilon^\alpha \rho^{\alpha_j} \hat{\mathbf{i}}^{\alpha_j} - \sum_{\beta \neq \alpha} \varepsilon^\alpha \rho^{\alpha_j} \hat{\mathbf{T}}_\beta^{\alpha_j} - \varepsilon^\alpha \rho^{\alpha_j} \nabla \tilde{A}^{\alpha_j} + \right. \\
& + \frac{1}{\rho^{\alpha_j}} \mathbf{E}_T \cdot \mathbf{P}^{\alpha_j} \nabla(\varepsilon^\alpha \rho^{\alpha_j}) - \varepsilon^\alpha \nabla \mathbf{E}_T \cdot \mathbf{P}^{\alpha_j} + \frac{1}{2} \varepsilon_o \mathbf{E}_T \cdot \mathbf{E}^{\alpha_j} \nabla \varepsilon^\alpha + \\
& \left. + \lambda_\rho^{\alpha_j} \nabla(\varepsilon^\alpha \rho^{\alpha_j}) + \varepsilon^\alpha \rho^{\alpha_j} \lambda_{q_e}^{\alpha_j} \nabla z^{\alpha_j} - \varepsilon^\alpha q_e^{\alpha_j} \mathbf{E}_T \right] + \\
& + \sum_{\alpha=l,s} \varepsilon^\alpha \mathcal{J}^\alpha \cdot \mathbf{E}_T + \\
& + \sum_{\alpha=l,s} \sum_{j=1}^N \sum_{\beta \neq \alpha} \varepsilon^\alpha \rho^{\alpha_j} \hat{\mathbf{e}}_\beta^{\alpha_j} \left[-\tilde{A}^{\alpha_j} - \frac{1}{\rho^{\alpha_j}} \mathbf{E}_T \cdot \mathbf{P}^{\alpha_j} - \frac{1}{2} (\mathbf{v}^{\alpha_j,\alpha})^2 - \lambda_\rho^{\alpha_j} - \right. \\
& \left. - \tilde{A}^\alpha - \frac{1}{2} (\mathbf{v}^{\alpha,s})^2 \right] + \\
& + \sum_{\alpha=l,s} \sum_{j=1}^N \varepsilon^\alpha \rho^{\alpha_j} \hat{\mathbf{r}}^{\alpha_j} \left[-\tilde{A}^{\alpha_j} - \frac{1}{\rho^{\alpha_j}} \mathbf{E}_T \cdot \mathbf{P}^{\alpha_j} - \lambda_\rho^{\alpha_j} - \frac{1}{2} (\mathbf{v}^{\alpha_j,\alpha})^2 \right] + \\
& + \sum_{\alpha=l,s} \nabla \cdot \mathbf{D}^{\alpha_j} [\varepsilon^\alpha \lambda_D^{\alpha_j}] + \\
& + \sum_{\alpha=l,s} \sum_{j=1}^N \frac{D^s z^{\alpha_j}}{Dt} \left[-\varepsilon^\alpha \rho^\alpha \frac{\partial \tilde{A}^\alpha}{\partial z^{\alpha_j}} + \varepsilon^\alpha \rho^{\alpha_j} \lambda_{q_e}^{\alpha_j} + \varepsilon^\alpha \rho^{\alpha_j} \Lambda \right] +
\end{aligned}$$

$$\begin{aligned}
& + \sum_{\alpha=l,s} \sum_{j=1}^N \nabla \times \mathbf{E}^{\alpha_j} \cdot [\mathbf{\Lambda}_E^{\alpha_j} \varepsilon^\alpha] + \\
& + \sum_{\alpha=l,s} \sum_{j=1}^N \lambda_D^{\alpha_j} \left[\mathbf{D}^{\alpha_j} \cdot \nabla \varepsilon^\alpha - \varepsilon^\alpha q_e^{\alpha_j} - \varepsilon^\alpha \widehat{d}^{\alpha_j} - \sum_{\beta \neq \alpha} \varepsilon^\alpha \widehat{d}_\beta^{\alpha_j} \right] + \\
& + \sum_{\alpha=l,s} \sum_{j=1}^N \mathbf{\Lambda}_E^{\alpha_j} \cdot \left[\nabla \varepsilon^\alpha \times \mathbf{E}^{\alpha_j} - \varepsilon^\alpha \widehat{\sigma}^{\alpha_j} - \sum_{\beta \neq \alpha} \varepsilon^\alpha \widehat{\sigma}_\beta^{\alpha_j} \right] + \\
& + \sum_{\alpha=l,s} \sum_{j=1}^N \lambda_{q_e}^{\alpha_j} \left[\nabla \cdot (\varepsilon^\alpha \mathcal{J}^{\alpha_j}) - \varepsilon^\alpha \widehat{q}^{\alpha_j} - \sum_{\beta \neq \alpha} \varepsilon^\alpha \rho^{\alpha_j} \widehat{Z}_\beta^{\alpha_j} \right] \geq 0,
\end{aligned}$$

where subscripted sym and as mean the symmetric and anti-symmetric part of the tensor, respectively.

Acknowledgements

J.H. Cushman would like to acknowledge the ARO/Terrestrial Sciences and Mathematics divisions for support under grant DAAG55-98-1-0228.

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